

ECS401: Bornat Test

FEEDBACK: Answers and Explanations

Actively work through these again, going through all the working again but trying to work out why the answer is what it is. If you still get the wrong answer then either you were just not careful enough and made a silly slip, or you have a faulty understanding of how assignment works, or you just don't understand assignment at all yet. In either case it is possible to cure the problem if you work at it. Get help from the lab and peer tutors.

1. What are the new values of a and b after the following code fragment has executed?

```
int a = 10; // L1
int b = 20; // L2
a = b;      // L3
```

Line	a	b
L1	10	
L2		20
L3	20	

a is now: b is now:

EXPLANATION:

We start with 10 in a and 20 in b. The assignment is read “a gets a copy of the value in b” so a copy of what is in b (20) is put into a, leaving b alone.

2. What are the new values of a and b after the following code fragment has executed?

```
int a = 10; // L1
int b = 20; // L2
b = a;      // L3
```

Line	a	b
L1	10	
L2		20
L3		10

a is now: b is now:

EXPLANATION:

We start with 10 in a, and 20 in b. The assignment is read “b gets a copy of the value in a” so a copy of what is in a (10) is put into b, leaving a alone.

3. What are the new values of big and small after the following code fragment has executed?

```
int big = 10; // L1
int small = 20; // L2
big = small; // L3
```

Line	big	small
L1	10	
L2		20
L3	20	

big is now: small is now:

EXPLANATION:

We start with 10 in `big`, and 20 in `small`. The assignment is read “`big` gets a copy of the value in `small`” so a copy of what is in `small` (20) is put into `big`, leaving `small` alone.

4. What are the new values of `a` and `b` after the following code fragment has executed?

```
int a = 10; // L1
int b = 20; // L2
a = b;      // L3
b = a;      // L4
```

Line	a	b
L1	10	
L2		20
L3	20	
L4		20

`a` is now: 20 `b` is now: 20

EXPLANATION:

We start with 10 in `a` and 20 in `b`. The first assignment is read “`a` gets a copy of the value in `b`” so a copy of what is in `b` (20) is put into `a`, leaving `b` alone. Both now hold the value 20. The second assignment is read “`b` gets a copy of the value in `a`” so a copy of what is in `a` (now 20) is put into `b`. That just means that the 20 that was already there is deleted and replaced with another copy of the same thing – 20.

5. What are the new values of `a` and `b` after the following code fragment has executed?

```
int a = 10; // L1
int b = 20; // L2
b = a;      // L3
a = b;      // L4
```

Line	a	b
L1	10	
L2		20
L3		10
L4	10	

`a` is now: 10 `b` is now: 10

EXPLANATION:

We start with 10 in `a` and 20 in `b`. The first assignment is read “`b` gets a copy of the value in `a`” so a copy of what is in `a` (10) is put into `b`, leaving `a` alone. Both now hold the value 10. The second assignment is read “`a` gets a copy of the value in `b`” so a copy of what is in `b` (now 10) is put into `a`. That just means that the 10 that was already there is deleted and replaced with another copy of the same thing – 10.

6. What are the new values of a, b and c after the following code fragment has executed?

```
int a = 10; // L1
int b = 20; // L2
int c = 30; // L3

a = b;      // L4
b = c;      // L5
```

Line	a	b	c
L1	10		
L2		20	
L3			30
L4	20		
L5		30	

a is now: b is now: c is now:

EXPLANATION:

Variable a starts holding value 10, b holding value 20 and c holding value 30. The first assignment puts a copy of what is in b into a, so both a and b end up holding 20. The second assignment puts a copy of what was in c into b. That overwrites the value of 20 that was in b with a copy of the 30 from c, with c unchanged (still 30). So a ends up with 20, and b and c with 30.

7. What are the new values of a, b and c after the following code fragment has executed?

```
int a = 5; // L1
int b = 3; // L2
int c = 7; // L3

a = c;      // L4
b = a;      // L5
c = b;      // L6
```

Line	a	b	c
L1	5		
L2		3	
L3			7
L4	7		
L5		7	
L6			7

a is now: b is now: c is now:

EXPLANATION:

Variable a starts with value 5, b with value 3 and c with value 7. The first assignment puts a copy of what is in c into a, so both a and c end up holding 7. The second assignment puts a copy of what is in a into b. That overwrites the value of 3 that was in b with another copy of the 7 from a, with a unchanged (still 7). At this point a has value 7, b has value 7 and c has value 7. The final line puts a copy of what is in b (7) into c. That overwrites the 7 that was there with a new value 7, so all three end up holding 7.

8. What are the new values of a, b and c after the following code fragment has executed?

```
int a = 5; // L1
int b = 3; // L2
int c = 7; // L3

c = b;      // L4
b = a;      // L5
a = c;      // L6
```

Line	a	b	c
L1	5		
L2		3	
L3			7
L4			3
L5		5	
L6	3		

a is now: b is now: c is now:

EXPLANATION:

Variable a starts with value 5, b with value 3 and c with value 7. The first assignment puts a copy of what is in b into c, so both b and c now hold 3, with c not touched so it is still 7. The second assignment puts a copy of what is in a into b. That puts a 5 into b, so at this point a is 5, b is 5 and c is 3. The final assignment puts a copy of what is in c into a, overwriting what was there. That means a gets the value 3.

9. What are the new values of a, b and c after the following code fragment has executed?

```
int a = 5; // L1
int b = 3; // L2
int c = 7; // L3

c = b;      // L4
a = c;      // L5
b = a;      // L6
```

Line	a	b	c
L1	5		
L2		3	
L3			7
L4			3
L5	3		
L6		3	

a is now: b is now: c is now:

EXPLANATION:

Variable a starts with value 5, b with value 3 and c with value 7. The first assignment copies the 3 from b into c, so both b and c now hold 3. The second assignment puts a copy of what is now in c (the 3 just put there) into a. That means a also now holds 3. The final assignment puts a copy of the 3 from a into b. All three variables end up holding the value 3.

10. What are the new values of a, b and c after the following code fragment has executed?

```
int a = 5; // L1
int b = 3; // L2
int c = 7; // L3

b = a;      // L4
c = b;      // L5
a = c;      // L6
```

Line	a	b	c
L1	5		
L2		3	
L3			7
L4		5	
L5			5
L6	5		

a is now: b is now: c is now:

EXPLANATION:

Variable a starts with value 5, b with value 3 and c with value 7. The first assignment copies the 5 from a into b, so both b and a now hold 5. The second assignment puts a copy of what is now in b (the 5 just put there) into c. That means c also now holds 5. The final assignment puts a copy of the 5 from c into a. All three variables end up holding the value 5.

11. What are the new values of a, b and c after the following code fragment has executed?

```
int a = 5; // L1
int b = 3; // L2
int c = 7; // L3

b = a;      // L4
a = c;      // L5
c = b;      // L6
```

Line	a	b	c
L1	5		
L2		3	
L3			7
L4		5	
L5	7		
L6			5

a is now: b is now: c is now:

EXPLANATION:

Variable a starts holding value 5, b holds value 3 and c holds value 7. The first assignment copies the 5 from a into b, so both b and a now hold 5. The second assignment puts a copy of what is in c (7) into a. That means a and c now hold 7 and b holds 5. The final assignment puts a copy of the 5 from b into c. b and c are left with 5 and a holds the 7 put there in the earlier step.

12. What are the new values of a, b and c after the following code fragment has executed?

```
int a = 5; // L1
int b = 3; // L2
int c = 7; // L3

a = c;      // L4
c = b;      // L5
c = a;      // L6
```

Line	a	b	c
L1	5		
L2		3	
L3			7
L4	7		
L5			3
L6		7	

a is now: b is now: c is now:

EXPLANATION:

Variable a starts holding value 5, b holds value 3 and c holds value 7. The first assignment copies the 7 from c into a, so both a and c now hold 7. The second assignment puts a copy of what is in b (3) into c. That means a holds 7 from the last step, b is so far unchanged holding 3 as there has been no assignment to it, and c now also holds 3. The final assignment copies what is now in a (7) into b, leaving b holding 7 as well as a, and c still with the 3 from the previous step.